

ARCHIMEDES Preparing REDCap Data for Submission

A guide for research teams sharing REDCap data with ARCHIMEDES

This guide explains what to send when sharing REDCap data with ARCHIMEDES. If anything here does not fit your study, contact the ARCHIMEDES technical support team before uploading. We are happy to help.

You only need to provide two files

Your REDCap **data dictionary** and your REDCap **data export**, with a few participant columns added. Everything else is handled on our side.

1. Where to put your files

```
/project_name/
├── deidentified-raw/
│   ├── clinical/
│   │   └── STUDY_redcap_data.csv          (your data export)
├── documentation/
│   └── data_dictionary/
│       └── STUDY_DataDictionary.csv      (your data dictionary)
```

Put everything inside one top folder named after your project, for example CNMDP/, and share that folder with us. The project name is how we identify which project the files belong to.

- Save both files as .csv, not .xlsx.
- Use file names without spaces.
- One export with all your forms, or separate exports for some forms, both work.

2. The data dictionary

In REDCap, go to **Project Setup**, then **Data Dictionary**, and download the current data dictionary. Send it unchanged: a file downloaded straight from REDCap already meets the rules below.

If you built or edited the dictionary yourself, check that:

- The column names are on the first line, with no title rows above them.
- The columns are in the same order as a REDCap download, as in the example below.
- Every row has a **Form Name**.
- Every **Field Type** is one of these, in lowercase with no extra spaces: text, notes, radio, dropdown, checkbox, yesno, calc, slider, file, descriptive, sql.
- Participant details (dob, sex, project, site, cohort) do not need to be listed as fields. The study_id row that REDCap includes can stay.

Example data dictionary (columns A to J)

	A	B	C	D	E	F	G	H	I	J
1	Variable / Field Name	Form Name	Section Header	Field Type	Field Label	Choices, Calculations, OR Slider Labels	Field Note	Text Validation Type OR Show Slider Number	Text Validation Min	Text Validation Max
2	moca_naming	moca		text	MoCA naming score			integer	0	3
3	moca_total	moca		text	MoCA total score			integer	0	30
4	phq9_q1	phq9		radio	Little interest or pleasure in doing things	0, Not at all 1, Several days 2, More than half the days 3, Nearly every day				
5	phq9_notes	phq9		notes	Assessor notes					

After column J, a REDCap dictionary continues with: Identifier?, Branching Logic (Show field only if...), Required Field?, Custom Alignment, Question Number (surveys only), Matrix Group Name, Matrix Ranking?, Field Annotation.

If you later add or change forms or fields in REDCap, let us know before sending new data.

3. The data export

In REDCap, go to **Data Exports, Reports, and Stats**, export all data, and choose **CSV / Microsoft Excel (raw data)**. Then check or add these columns:

Column	Needed	What to enter	Example
study_id	Yes	Your participant ID	P001
redcap_event_name	Yes	Already in your export	visit_1_arm_1
dob	Yes	Date of birth as YYYY-MM-DD, YYYY-MM or YYYY	1985-03
sex	Yes	Male, Female or Other	Female
project	Preferred	Your project name. If not sure, check with the ARCHIMEDES technical support team, who can send you the correct name	CNMDP
site	Preferred	Your site name. If not sure, check with the ARCHIMEDES technical support team, who can send you the correct name	University of Ottawa Heart Institute
cohort	Preferred	The participant's group. If not sure, check with the ARCHIMEDES technical support team, who can send you the correct name	Affected

Example data export

	A	B	C	D	E	F	G	H	I	J	K
1	study_id	redcap_event_name	project	site	cohort	dob	sex	moca_total	moca_complete	phq9_q1	phq9_complete
2	P001	visit_1_arm_1	CNMDP	University of Ottawa Heart Institute	Affected	1985-03	Female	26	2	1	2
3	P001	visit_2_arm_1	CNMDP	University of Ottawa Heart Institute	Affected	1985-03	Female	27	2	0	2
4	P002	visit_1_arm_1	CNMDP	University of Ottawa Heart Institute	Control	1971-08	Male	22	2	2	2

This example holds two forms, moca and phq9, matching the data dictionary example above. The column names must be on the first line, with no title rows above them. Keep all other columns exactly as REDCap exported them. Use lowercase column names as shown.

4. Simple rules

Rule	Good	Avoid
Participant IDs use letters and numbers only	P001	P_001, P-001
Same ID for a participant on every row	P001, P001	P001, P1
Dates as YYYY-MM-DD, YYYY-MM or YYYY only	1985-03-22, 1985-03, 1985	Any other format
Sex written in full	Male, Female, Other	M, F, 1, 2
Date of birth and sex on every row for the participant	Filled on each visit row	Only on the first visit
No names or identifying information in IDs	P001	SmithJ
Project, cohort, site and event names exactly as on the list we send you and configured on ARCHIMEDES platform	Same spelling, capitals and spacing	Extra spaces, different capitals

Once your project is set up, the ARCHIMEDES technical support team sends you the list of accepted project, cohort and event names.

Only the year and month of birth are kept, so send whatever date precision you are comfortable with.

5. Before you send

Remove identifying information

- Names, initials, contact details, addresses and health or medical record numbers
- Free-text notes and comments
- Dates that are not needed, such as consent or appointment dates

Good to know

- Tell us your visit (event) names and forms before your first upload, so we can set them up.
- Send data once data entry is final. To correct data we already have, contact the ARCHIMEDES technical support team.

- To send more data later, upload a new full export with the same file name.
- If your project uses checkboxes, calculated fields or repeating instruments, let us know first.
- If something in a file cannot be loaded, we will contact you.

6. Common mistakes to avoid

What to avoid: a title row above the column names, and values in the wrong format (shaded)

	A	B	C	D
1	Data_ingestion_test_V1			
2	study_id	redcap_event_name	dob	sex
3	P001	visit_1_arm_1	5/23/2024	F
4	P002	visit_1_arm_1	2024-05	M

These are the issues we see most often in files shared with us. Fixing them before you send saves a round of corrections.

Avoid	Example	Instead
Title rows above the column names, in the data dictionary or the data file	Data_Dictionary_V1	Column names on the first line
Blank Form Name in the data dictionary	Left empty	A Form Name on every row
Field Type with capital letters	Text	Lowercase: text
Extra spaces before or after a value. The cell looks the same, so check carefully	"text "	No extra spaces
Sex as a letter or code	F, M	Female, Male or Other
Date of birth in another format	5/23/2024	2024-05-23, 2024-05 or 2024
Project, cohort, site or event names that differ from the accepted list	Different spelling or extra spaces	Exactly as on the list we send you

7. Checklist

Files ☐ Data dictionary in documentation/data_dictionary/ ☐ Raw data export in deidentified-raw/clinical/ ☐ Both saved as .csv ☐ Column names on the first line, no title rows	Data ☐ study_id, dob and sex on every row ☐ project, site and cohort filled; if not sure, check with support ☐ Dates as YYYY-MM-DD, YYYY-MM or YYYY; sex as Male, Female or Other ☐ Names match the accepted values list ☐ Identifying information removed
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